



LAB MANUAL

CHY1005: Introduction to Computational Chemistry



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Introduction to Computational Chemistry

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Duration: 90 min.

Practical 1: Molecular Visualization

Aim

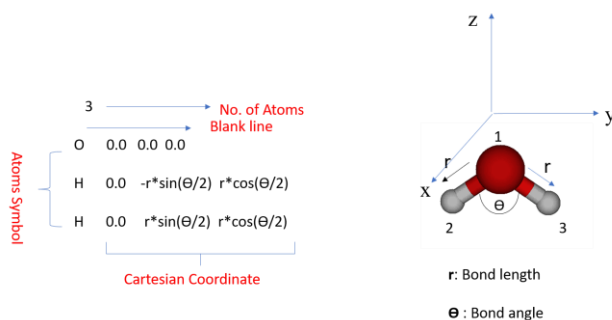
To build and visualize molecules using Avogadro.

Theory:

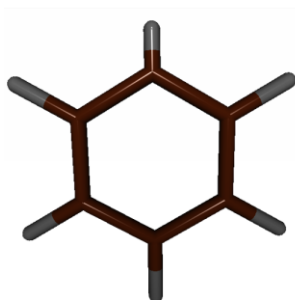
There are many tools to visualize the molecule. In this session, we will be using Avogadro, an open-source software. Generally, computational chemistry allows chemists to analyze properties of molecules using a computer. The computer can predict quantities such as: molecular geometry (bond length, bond angle), molecular orbitals, bonding, spectroscopy, and energy. It can also simulate the motion of molecules (molecular dynamics). The first step for these calculations is building the molecule. Building a molecule in computer involves describe the molecule to the computer and tell it what you want it to calculate. If you wanted to calculate the bond length in carbon dioxide you would specify an initial guess for the (x,y,z) coordinates of the three atoms. Then you would add some keywords to tell the computer to find the optimum geometry. This can be accomplished using one of the dedicated software. The generated structures can be saved in either of the following two formats:

1. Cartesian format
2. Z-matrix

Cartesian Geometry of Water:



Z-matrix of Geometry of Benzene:



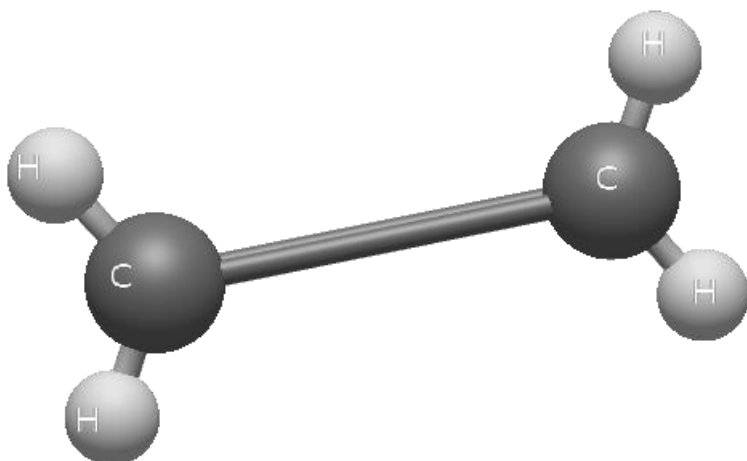
1	C						
2	C	1	1.40				
3	C	2	1.40	1	120		
4	C	3	1.40	2	120	1	0.0
5	C	4	1.40	3	120	2	0.0
6	C	5	1.40	4	120	3	0.0
7	H	2	1.07	3	120	4	180
8	H	3	1.07	2	120	1	180
9	H	4	1.07	3	120	2	180
10	H	5	1.07	4	120	3	180
11	H	6	1.07	5	120	4	180
12	H	1	1.07	2	120	3	180

Procedure:

- Molecules can be built using the Draw Tool of Avogadro GUI window.
- Open Avogadro, select the Draw Tool.
- Select the Draw tool (at the left end of the toolbar; looks like a pencil) to start building a molecule. In the left pane, make sure that the Adjust Hydrogens checkbox is checked.
- Select carbon from the dropdown menu. Click once in the drawing window (the black field). To create a bond to a second carbon, click the first carbon and drag away from it.
- If you want to delete an atom, you can right-click it while in drawing mode. If you want to add another type of atom, you can select it from the dropdown menu (select "Other..." and choose any element from the periodic table)
- Single click puts an isolated atom, eventually with its hydrogen atoms. If you want to create a bond, drag until the bond is long enough and release the mouse button.
- Bond order can be changed through the Bond Order drop down menu, or by typing the numbers "1", "2", or "3". Bonds can also be added to a molecule by left clicking on a bond that has already been created (this process cycles through single, double, and triple bonds). Right clicking on a bond will delete the bond, and the atom it's bonded to.
- Click "File" --> "Save" to save the molecule for later use.

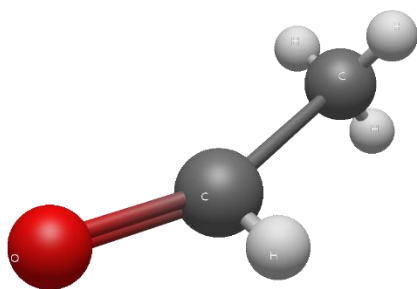
Results:

1) Ethelene:



H	1.472750	-0.146547	-1.448257
H	1.942969	0.265969	0.296284
H	-2.730272	1.552471	-0.717135
H	-2.260052	1.964987	1.027406
C	-2.005482	1.560782	0.069955
C	1.218180	0.257658	-0.490806

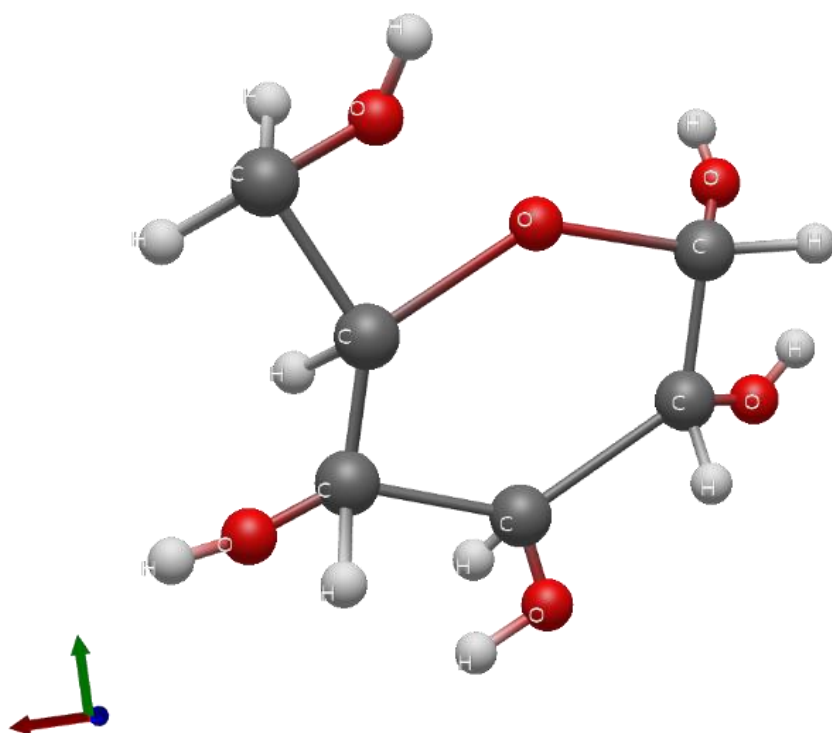
2) Acetaldehyde:



C	-2.262810	0.411420	-0.000000
C	0.561533	0.522615	0.000000
O	2.261444	0.719514	-0.872805
H	0.820276	0.379664	1.028356
H	-2.625041	0.545749	-0.997820
H	-2.582285	-0.540260	0.370326
H	-2.650275	1.186677	0.627493

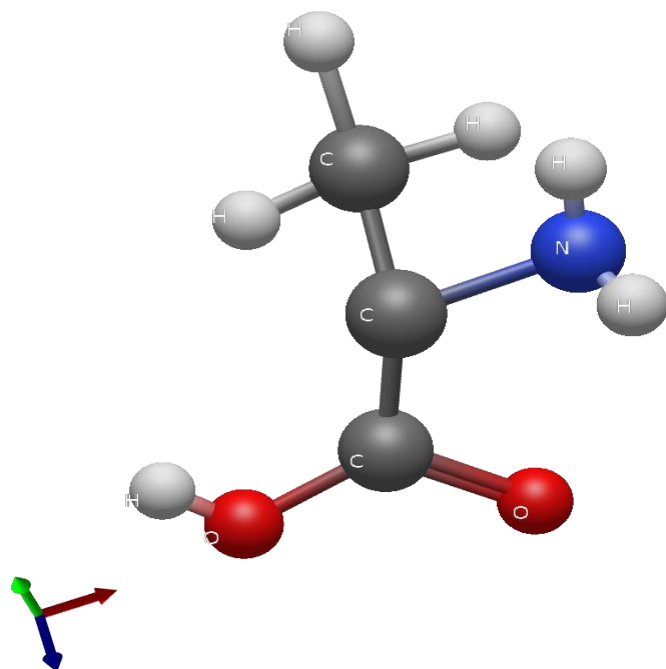


3) Glucose:



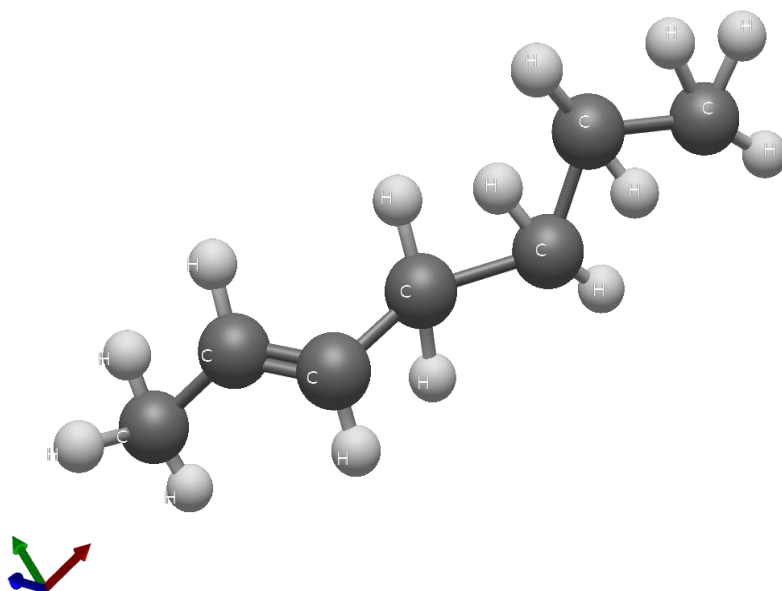
C	2.200016	3.508365	-1.849335
C	1.403744	1.025286	-0.910566
C	2.166800	-1.244603	-0.777713
C	-0.122080	-2.220995	0.552591
C	-2.978546	-0.678461	1.599647
C	-3.744830	1.822851	1.416973
O	-1.302891	2.523175	0.073516
O	0.729682	4.438107	-0.931853
H	-0.224670	5.442246	-1.332355
H	4.174981	2.910160	-1.751373
H	2.601702	5.006431	-0.859763
H	3.207984	0.702478	-0.236718
H	1.860043	-2.751801	-1.727632
H	0.905414	-2.870730	0.687189
H	-5.554173	1.745350	2.305149
O	-0.269903	-3.806322	0.810552
O	3.596656	-1.850027	-1.401332
O	-3.992994	-0.827091	2.577663
H	-3.340759	-2.149647	1.261888
H	4.371838	-2.015150	-2.213529
H	0.937032	-4.405633	0.383352
H	-4.862706	-0.024759	2.705636
O	-3.902473	3.100321	2.266196
H	-3.639097	4.177202	2.515496

4) Alanine:



```
H 1.334069 1.070051 0.744149
C -1.107973 1.349814 -0.000000
C -2.155475 -0.112761 0.792765
N 0.911638 1.194543 -0.186955
H 1.283193 2.051686 -0.620721
H -4.482000 0.507068 0.809300
O -1.094982 -1.527336 1.295673
O -4.040374 -0.279330 1.107702
C -0.958502 2.247322 -1.587296
H 0.420510 2.016438 -1.752639
H -2.256664 2.291758 -1.314149
H -0.647783 3.434735 -2.466798
```

5) 2-Heptene:



```
H 4.125329 -0.384932 -2.756483
C -1.264985 0.674705 0.260297
C -0.191692 -0.469677 -0.299695
C 1.136364 0.008020 -1.124487
C 2.717563 -0.393372 -2.167772
C 4.281576 0.446601 -2.056475
C 5.455578 0.082566 -3.416063
H 2.912767 0.415673 -1.376529
H -3.367753 0.527693 1.498230
H 5.658084 -0.443283 -4.506443
H -0.842452 -1.297843 -0.820058
H 0.186485 -0.712304 -1.778758
H 5.901622 0.752379 -2.473818
H 2.689392 -0.861270 -3.179917
H -2.591937 1.575379 0.952767
H 4.520442 0.935403 -0.976047
H 6.651155 0.496569 -3.259707
C -3.121646 0.728925 0.476546
H -3.575584 -0.008793 -0.151636
H 1.764396 1.051615 -0.676395
H -0.704053 1.732611 0.613009
```